

	Horizon Campus
	Faculty of Information Technology
	BSc (Hons.) in Information Technology/BIT (Hons.) in Networking and Mobile Computing

Please fully complete all sections as requested.

Degree Program: BSc (Hons.) in Information Technology		
BIT (Hons.) in Networking and Mobile Computing		
Research Details		
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Proposed Research Project Details:		
1. Working Title		
Augmented Reality–Based Interior Design Mobile Application for Wall & Floor Color Visualization with Personalized Color Palette Recommendation		
2. Problem Statement		
Interior design decisions, especially regarding wall and floor colors, are often challenging for homeowners and designers because colors look different in reality than in catalogues or design boards. Customers usually struggle to imagine how a selected color would appear in		

their actual space, leading to costly mistakes and dissatisfaction. Traditional 2D color samples do not provide realistic visualization, while hiring professional designers can be expensive. At the same time, color harmony and personalized design preferences are rarely addressed by existing tools. Therefore, there is a need for a mobile application that leverages **Augmented Reality (AR)** to allow real-time visualization of wall and floor colors, while also providing intelligent **color palette recommendations** based on user preferences. Such a solution would save time, reduce cost, and improve customer satisfaction by enabling data-driven and personalized design decisions.

3. Research Aim

To design and develop an **Augmented Reality–based mobile application** that enables real-time visualization of wall and floor colors and recommends personalized color palettes based on user preferences, enhancing decision-making in interior design.

4. Research Objectives

Objective 1: To implement an AR-based feature that allows users to visualize selected wall and floor colors in real-time using their smartphone camera.

Objective 2: To design a color recommendation system that generates personalized color palettes based on user input and established design principles.

5. Research Questions

Question 1: How can AR technology be applied to accurately visualize wall and floor colors in real-world environments?

Question 2: What methods can be used to generate personalized and aesthetically pleasing color palettes based on user input?

Question 3: To what extent does the proposed app improve decision-making and satisfaction compared to traditional color selection methods?

6. Background Literature Review

Interior design applications have increasingly incorporated technology to improve visualization and decision-making. **Augmented Reality (AR)** has been widely studied for

its ability to project virtual elements into physical spaces. Houzz and IKEA's AR apps demonstrate how users can preview furniture and colors in real environments, though these are limited in personalization. Research by Emersya (IEEE Spectrum, 2022) highlights how AR enables consumers to customize products, reinforcing the need for similar solutions in interior design.

Wall and floor color visualization remains a key challenge. According to *Automatic Interior Design in AR Based on Procedural Rules* (ResearchGate, 2021), AR can automatically adapt interior layouts, while *Background-Aware Colorization in AR* (ResearchGate, 2022) emphasizes the importance of harmonizing colors with environmental factors.

Personalization in interior design has been explored in studies like *Interactive Interior Design Recommendation via Multimodal Reinforcement Learning* (ArXiv, 2023), which uses AI to align design with user feedback. Similarly, *Aesthetic Preference Prediction Using Fuzzy Logic* (ArXiv, 2024) demonstrates how computational models can predict human color preferences.

Color harmonization is another focus area. *Color Harmonization in AR Applications* (MDPI, 2021) discusses how visual balance impacts user experience, directly aligning with this project's recommendation system.

Despite these advances, most existing solutions treat visualization and recommendation as separate problems. Few combine **AR visualization with intelligent personalization**, leaving a gap for a system that both shows how colors look and recommends palettes tailored to user needs. This project fills that gap by integrating AR visualization with an AI-driven recommendation system, offering a comprehensive solution.

Thus, the research builds upon existing AR visualization and AI-based recommendation systems but innovates by combining them into a unified, user-friendly mobile app for interior design.

7. Data Requirement

Primary Data

1. User Surveys (Example: Google Forms or in-app questionnaire)

- Surveys can ask users about their favorite colors, preferred wall–floor combinations, lighting conditions at home, and overall design preferences.
- Why: This helps the recommendation system learn real user preferences instead of relying only on theory.

2. User Testing of the Prototype (Example: Asking 10–15 users to try the app and give feedback)

- Users interact with the AR prototype, test wall and floor color visualization, and then rate usability, realism, and satisfaction.
- Why: This evaluates whether the app is user-friendly and if the recommendations feel practical and realistic.

Secondary Data

1. Existing Datasets on Color Theory and Palette Combinations (Example: Adobe Color CC database, COCO color dataset)

- These provide predefined harmonious color palettes (complementary, triadic, analogous, etc.).
- *Why*: The recommendation system can suggest scientifically proven color matches instead of random combinations.

2. Published Research on AR Visualization Techniques (Example: IEEE papers on AR interior design)

- Previous studies give insights into best practices for AR rendering, environmental lighting effects, and realism in visualization.
- *Why*: Helps improve the accuracy and performance of the AR module.

8. Hardware/ Software Requirements

Hardware:

Android smartphone with ARCore support

Software:

- Flutter SDK:
Free and open source.
- Android Studio for mobile app development.
Free to download.
- ARCore SDK for AR visualization.
Free and officially supported by Google and Apple.
- Python/ML libraries (TensorFlow, scikit-learn) for recommendation system.
- Firebase for backend and data storage.
Free tier available

9. Ethics

This project involves **human participants** for usability testing and surveys. Ethical considerations include:

- Informed consent from participants.
- Anonymity of collected data.
- Ensuring no harm or discomfort during user testing.
No animal, environmental, or political issues are involved.

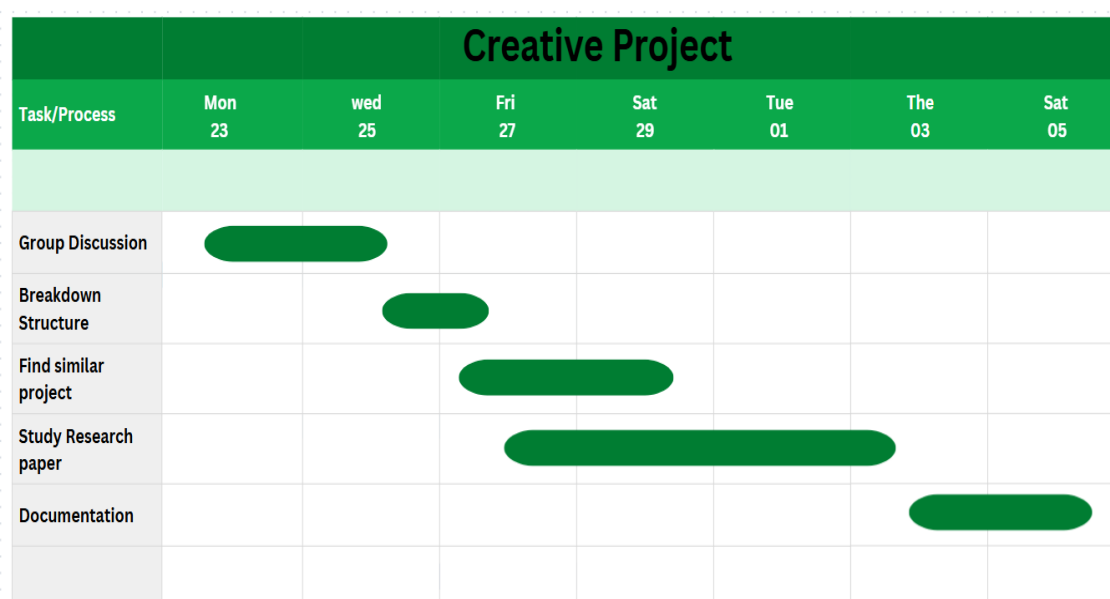
10. References

1. **R. S. P., H. A., and S. S.**, “Augmented Reality in Interior Design,” *Journal of Innovative Image Processing*, vol. 6, no. 3, pp. 305–313, Sep. 2024. DOI: 10.36548/jiip.2024.3.007.[Link](#)
2. **IKEA**, “Launch of new IKEA Place app,” *IKEA Newsroom*, Sep. 12, 2017. Describes ARKit-based AR visualization with 98% accuracy in scaling products and realistic light/shadow rendering.[Link](#)
3. **E. Gürçınar and Ö. C. Esen**, “The Application of Augmented Reality in Interior Design Education,” in *Proc. NordDesign (DS 91)*, Linköping, Sweden, Aug. 2018.

Explores AR usability in interior design education, with user feedback collected via surveys.[Link](#).

4. **U. Sridevi et al.**, “Integrating Augmented Reality and Machine Learning for Enhanced Interior and Exterior Design Visualization and Recommendation,” *J. Propulsion Technology*, vol. 45, no. 3, pp. 2415– , 2024. Presents AR + ML integration for real-time design previews and personalized suggestions.[Link](#)
5. **C. E. Mendoza-Ramírez et al.**, “Augmented Reality: Survey,” *Appl. Sci.*, vol. 13, no. 18, 2023. Offers a systematic literature review of AR technologies, challenges, and trends across multiple domains.[Link](#)
6. **IKEA Kreativ**, *IKEA Newsroom*, Jun. 22, 2022. Highlights IKEA’s AI-powered, spatial computing and mixed-reality tools for lifelike room design experiences.[Link](#)
7. **Houzz**, *Wikipedia*, latest update. Describes Houzz's "View in My Room 3D" AR feature allowing users to place virtual products in real spaces using iOS and ARCore.[Link](#)
8. **Revathy et al. (2024)**: Demonstrates real-time AR visualization applied to interior furniture placement using web-based AR.
9. **Sridevi et al. (2024)**: Integrates AR and ML for immersive visualization and user-tailored recommendations.

11. Gantt Chart



Students Agreement:

By signing below on 2025/07/06, I/We, as J.M.V.Sandaru , M.T.Upeksha, (name/s) agree to the commonly known and accepted research ethics and guidelines. I/We commit to conducting the research project as mentioned above and outlined in the agreement. I/We understand that any changes to the group composition, identified research problem, or selected supervisor may result in penalties.

Name with Initials**Registration No****Signature**

J.M.V.Sandaru

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M.T.Upeksha

ITBIN-2211-0307

**Supervisor(s) Agreement:**

By signing below, I/We agree to supervise the aforementioned group of students throughout their research project, providing guidance and support for successful completion. Furthermore, by signing this document, I/we give my/our consent to the following:

1. The identified research problem is justifiable and feasible for investigation.
2. The research aim is achievable within a one-year timeframe.
3. The specified research objectives are measurable and achievable within the research period.

Supervisor's Name with Initials**Signature****Date**

Mr.Anjana Abeykoon

Departmental Use Only:**Date Proposal Form Submitted:** _____

All required sections of the form have been accurately completed, and the necessary signatures have been placed. The initial research proposal has been partially approved, thereby allowing the proposed research project to proceed in accordance with the specified guidelines outlined in the IT41028.

Research Coordinator's Signature: _____ Date: ____

Department Head's Signature: _____ Date: ____